

Day 12 Activity: A flights investigation

Wednesday, June 3, 2026

Your Name Here

Setup

```
library(tidyverse)
library(nycflights13)
```

Today we use `flights`, every flight out of NYC in 2013. Glimpse it, and note the mess: cancelled flights have **missing** `dep_delay` / `arr_delay`.

```
glimpse(flights)
```

Rows: 336,776

Columns: 19

```
$ year      <int> 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2013, 2~
$ month     <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ day       <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ dep_time  <int> 517, 533, 542, 544, 554, 554, 555, 557, 557, 558, 558, ~
$ sched_dep_time <int> 515, 529, 540, 545, 600, 558, 600, 600, 600, 600, 600, ~
$ dep_delay <dbl> 2, 4, 2, -1, -6, -4, -5, -3, -3, -2, -2, -2, -2, -2, -
1~
$ arr_time  <int> 830, 850, 923, 1004, 812, 740, 913, 709, 838, 753, 849,~
$ sched_arr_time <int> 819, 830, 850, 1022, 837, 728, 854, 723, 846, 745, 851,~
$ arr_delay <dbl> 11, 20, 33, -18, -25, 12, 19, -14, -8, 8, -2, -3, 7, -
1~
$ carrier   <chr> "UA", "UA", "AA", "B6", "DL", "UA", "B6", "EV", "B6", "~
$ flight    <int> 1545, 1714, 1141, 725, 461, 1696, 507, 5708, 79, 301, 4~
$ tailnum   <chr> "N14228", "N24211", "N619AA", "N804JB", "N668DN", "N394~
$ origin    <chr> "EWR", "LGA", "JFK", "JFK", "LGA", "EWR", "EWR", "LGA",~
$ dest      <chr> "IAH", "IAH", "MIA", "BQN", "ATL", "ORD", "FLL", "IAD",~
$ air_time  <dbl> 227, 227, 160, 183, 116, 150, 158, 53, 140, 138, 149, 1~
```

```
$ distance      <dbl> 1400, 1416, 1089, 1576, 762, 719, 1065, 229, 944, 733, ~
$ hour          <dbl> 5, 5, 5, 5, 6, 5, 6, 6, 6, 6, 6, 6, 6, 6, 5, 6, 6, 6~
$ minute        <dbl> 15, 29, 40, 45, 0, 58, 0, 0, 0, 0, 0, 0, 0, 0, 59, 0~
$ time_hour     <dtm> 2013-01-01 05:00:00, 2013-01-01 05:00:00, 2013-01-
01 0~
```

No new verbs today. You have all six (`filter`, `arrange`, `select`, `mutate`, `group_by`, `summarise`), plus `count()` and the pipe. The goal is **sequencing them and checking your work**.

Part 1 — Plan before you code

For this question: *Which of the three NYC airports (*origin*) sent out the most flights in 2013?*

1a. First, write the **English plan**, one sentence per step. Which verbs, in what order? What does one row of your target table represent?

[Your plan here.]

1b. Now write it. (Hint: this is a one-liner with `count()`, or a `group_by()` + `summarise()`.)

```
# Your code:
```

Part 2 — Build a pipeline

Which carrier had the worst average arrival delay, among carriers with at least 1,000 flights?

Build it step by step: `filter` is **not** needed on the raw rows here, but you'll need it on the **summary table** to drop small carriers. Include `n = n()`.

```
# Your code:
```

Which carrier is it, and how many flights is that average based on?

[Your answer. We only have the 2-letter code today; turning it into the airline's name needs a *join*, which is coming later.]

Part 3 — mutate(), then plot

3a. Create a label `time_of_day` that is "morning" when `dep_time < 1200` and "afternoon/evening" otherwise. Then find the average departure delay for each, with a count.

```
# Your code:
```

3b. Pipe a version of that summary into a `geom_col()` plot of average delay by `time_of_day`. (Remember to switch `|>` to `+` at `ggplot().`)

```
# Your code:
```

One sentence: what does it show?

[Your answer.]

Part 4 — Don't get fooled

4a. Order matters. These two pipelines run without error but answer **different questions**. Run both, then explain in one or two sentences how the questions differ.

```
# Pipeline A
flights |>
  filter(dest == "BOS") |>
  summarise(avg = mean(arr_delay, na.rm = TRUE))
```

```
# Pipeline B
flights |>
  group_by(dest) |>
  summarise(avg = mean(arr_delay, na.rm = TRUE)) |>
  filter(dest == "BOS")
```

[How do the questions differ? Do the numbers match? Should they?]

4b. Spot the silent NA. This returns NA for some carriers. Why? What single change fixes it?

```
flights |>
  group_by(carrier) |>
  summarise(avg_delay = mean(dep_delay))
```

[Your explanation and fix.]

Part 5 — Open investigation

Pick **one** of these questions (or invent your own), and answer it with a pipeline:

- Which destination has the most *variable* arrival delays? (Hint: `sd()` is a reduction, like `mean()`.)
- Do flights later in the day make up time in the air more than morning flights?
- Which day of the week (you'd need `month/day...`) is worst to fly? (*Stick to month if day-of-week is too much.*)

5a. State your question and your English plan.

[Question + plan.]

5b. Your pipeline:

```
# Your code:
```

5c. Required: report your checks. Answer all three:

- How did you handle missing values?
- What's the smallest `n` in your result, and is it big enough to trust?
- Does the answer pass a smell test (does it match what you'd expect)?

[Your three checks.]

Wrap-up

One-sentence reflection: what's one mistake you almost made today (wrong order, forgotten `na.rm`, a tiny group) that a quick check caught?

[Write one sentence.]

Code appendix

```
library(tidyverse)
library(nycflights13)
glimpse(flights)
# Your code:

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# Pipeline A
flights |>
  filter(dest == "BOS") |>
  summarise(avg = mean(arr_delay, na.rm = TRUE))
# Pipeline B
flights |>
  group_by(dest) |>
  summarise(avg = mean(arr_delay, na.rm = TRUE)) |>
  filter(dest == "BOS")
flights |>
  group_by(carrier) |>
  summarise(avg_delay = mean(dep_delay))
# Your code:
```